

TECHLA.LABS

Techla.labs .learn

Product Requirements Document

An AI-powered learning platform built around one question: what should I learn next, and why?

Version	1.0 (draft)
Status	Ready for review
Date	September 19, 2026
First domain	Data Structures & Algorithms (DSA)

01 Overview

Techla.labs.learn is a learning platform that builds a personalized path for each learner instead of handing them a fixed course. It finds out what the learner already knows, maps what they are missing, and keeps answering one question: what should I learn next, and why?

The first domain is Data Structures & Algorithms (DSA). The architecture must support more domains later: Frontend, Backend, Databases, Full-Stack and Web Security.

CURRENT STATE

A Next.js + Supabase starter exists (landing page, domain selector, DSA path, prototype diagnostic, schema, RLS). It has not passed a production build, and the diagnostic and path engine are still demo-level. This PRD defines the real product.

02 Problem

- **Static courses waste time.** Everyone follows the same order, so learners repeat what they know and hit topics they are not ready for.
- **Hidden gaps.** Beginners cannot tell which prerequisite they are missing when a topic feels too hard.
- **Scattered tools.** Learners bounce between videos, roadmaps and problem sites with no view of their mastery.
- **Unexplained advice.** Lists like “Recommended for you” give no reason, so learners do not trust them.

03 Goals and Non-Goals

Goals

1. Assess real DSA knowledge per topic, not one overall score.
2. Generate a path from a prerequisite graph, with an explanation for every recommendation.
3. Pair each topic with free resources and practice problems.
4. Track mastery from evidence (assessments, practice, repetition, recency), not page views.
5. Make the core path work with no AI dependency.

Non-goals for v1

- Hosting video or original course content (link to free external resources).
- An in-app code editor or judge (link to LeetCode).
- Payments, social features and certificates.
- Domains other than DSA (only “Coming Soon” cards).

04 Target Users

Persona	Need
Self-taught beginner	A clear starting point and simple explanations
Interview candidate	The fastest route to their gaps, with a target date
CS student	Structure that complements coursework
Career switcher	Skip what they know, fill what they do not

05 Product Principles

- **Graph, not list.** Topics are nodes with prerequisites, never a linear syllabus.
- **Explain every recommendation.** Each “next step” carries its reason.
- **Mastery is earned.** Evidence counts. Opening a page does not.
- **Deterministic first.** AI enhances the experience but is never required for the path.
- **Learn while working.** Every topic pairs a concept with practice.

06 Scope

MVP (v1)	Later
• Landing page, /learn domain selector, /learn/dsa overview • Auth (email and magic link) • DSA diagnostic (50+ questions) with per-topic scoring • Personalized path engine • Dashboard, topic pages, resources, practice problems • Mastery tracking and persistence • Supabase schema, RLS and seed data	• AI tutor and AI code review • Adaptive difficulty • Learning goals: interview, university, competitive programming, career switch, general CS • Target date (“ready in 12 weeks”) that reshapes the path • Additional domains

07 Key User Flows

First-time learner Landing page → Get Started → magic-link sign in → /learn → choose DSA → diagnostic → result map → start recommended path.

Returning learner Dashboard → Continue Learning → topic page → study a resource → solve problems → check-in assessment → mastery updates → next recommendation.

08 Functional Requirements

ID	Feature	Requirement
FR-1	Landing (/)	Minimal hero-first page. Nav: TECHLA.LABS, Learn, Paths, About, Sign in, Get Started. Badge “AI-POWERED LEARNING”; headline “Stop following courses. Start learning what you actually need.”; CTAs Start Learning and Explore Paths; product preview showing Assess → Understand → Personalize → Practice → Master.
FR-2	Domain selector (/learn)	Heading “What do you want to master?” Six cards: DSA (Available); Frontend, Backend, Databases, Full-Stack, Web Security (Coming Soon).
FR-3	DSA overview (/learn/dsa)	Progress %, current level (default “Not assessed”), recommended next step, Start Diagnostic CTA and a knowledge graph preview.
FR-4	Curriculum and graph	40+ topics across Foundations, Arrays & Strings, Linked Lists, Stacks & Queues, Hashing, Trees, Graphs, Sorting, Searching, Greedy, DP, Backtracking and Advanced. Each has id, slug, title, description, difficulty, time, category, prerequisites, resources, problems and mastery criteria.
FR-5	Diagnostic	50+ questions covering Big-O through backtracking. Types: multiple choice, code understanding, output prediction, conceptual, problem strategy. Explanation after each answer. Highly usable on mobile.
FR-6	Scoring	Score per topic, not just correct over total. Store score, confidence, attempts and last-assessed date per user per topic.
FR-7	Result page	Counts of mastered, developing and needs-attention concepts; ordered recommended path; Start My Path CTA.
FR-8	Path engine	generatePersonalizedPath({ userId, domainId }) loads topics, prerequisites and mastery; classifies mastered, weak and locked topics; returns current topic, ordered recommendations, completed and locked topics, and reasoning per recommendation.
FR-9	Recommendation logic	A topic is available only when all prerequisites have mastery of 4 or higher. If a wanted topic has an unmet prerequisite, recommend the deepest unmet one first.
FR-10	Dashboard	Overall and DSA mastery, topics mastered, problems solved, streak, Continue Learning card, knowledge map with status per area, and a Recommended Next card with a “Why”.
FR-11	Topic page	Title, description, difficulty, time, prerequisites and progress. Sections: Learn, Free Resources, Practice (easy, medium, hard) and Check Your Understanding.
FR-12	Resources	Real, free, working links (YouTube, docs, articles), 30+ at launch. Provider-agnostic data model.
FR-13	Practice problems	30+ valid LeetCode links at launch, tagged by topic and difficulty, with per-user status (not started, attempted, solved).

ID	Feature	Requirement
FR-14	Mastery engine	Levels 0 to 5. Inputs: assessment results, practice results, repeated attempts, difficulty and recency. Opening a topic never grants mastery.
FR-15	Authentication	Supabase Auth (email and magic link) at /auth and /auth/callback. Guests can browse public content; saving progress requires sign-in.
FR-16	AI abstraction	An AIProvider interface (generateExplanation, generateHint, evaluateAnswer) not tied to any vendor. The path must work with AI disabled.

09 Knowledge Graph and Path Logic

DSA is modeled as a directed prerequisite graph. If the learner wants Graphs, needs BFS, and BFS needs Queue, the system recommends Queue first and explains why.

EXAMPLE PATH

Queue → Graph Representation → BFS → DFS → Graph problems

Mastery levels

Level	Name	Meaning
0	Unknown	No evidence of knowledge
1	Introduced	Has seen the topic
2	Learning	Studying, partial assessment results
3	Practicing	Solving problems, inconsistent
4	Proficient	Reliable results. Unlocks dependent topics
5	Mastered	Repeated strong performance over time

10 Data Model

Supabase (PostgreSQL) tables:

Table	Purpose
profiles	User profile linked to auth.users
learning_domains	DSA and future domains, with status
topics	Curriculum nodes (with parent_id)
topic_prerequisites	Directed edges of the graph

Table	Purpose
diagnostic_questions	Question bank per topic
diagnostic_attempts / answers	Each attempt and each answer
user_topic_mastery	Mastery level, score, confidence, attempts (unique per user and topic)
resources	Free learning links per topic
practice_problems	LeetCode problems per topic
user_problem_progress	Per-user problem status
learning_sessions	Study time tracking

Security (Row Level Security)

- Curriculum, resources and problems are publicly readable.
- Profile, mastery, attempts, answers and problem progress are readable and writable only by their owner.
- No user may ever read another user's learning data.
- Service-role keys are never exposed to the client.

11 Design System

Dark theme and orange brand only. No other hues.

Swatch	Token	Hex	Use
	Background	#050505	Page background
	Surface	#0B0B0B	Cards and panels
	Border	#262626	Dividers and outlines
	White	#FFFFFF	Headlines and key text
	Muted	#A1A1AA	Secondary text
	Orange (brand)	#FF6A00	Primary actions and emphasis
	Orange light	#FF8533	Hover and highlights

- Feel: premium, technical, minimal, developer-focused. Avoid bootstrap-style layouts, childish illustrations, heavy gradients and clutter.
- Buttons: primary is orange with black text; secondary is transparent with a thin gray border.
- Cards: near-black surface, subtle border, orange border glow on hover.
- Motion: subtle only (transitions, hover, progress bars, counters). Respect prefers-reduced-motion.
- Responsive on desktop, tablet and mobile with no horizontal overflow.

12 Technical Requirements

- Next.js (App Router), TypeScript, Tailwind CSS, React, Supabase (Auth and Postgres).
- Server Components by default; Client Components only for interaction.
- Reusable, composable components; no giant components.
- Loading, empty and error states throughout, with no raw database errors shown to users.
- Seed data: DSA domain, 40+ topics with prerequisites, 50+ questions, 30+ resources, 30+ problems.
- README (setup, migrations, seeding, architecture) and .env.example with the public Supabase variables.

13 Success Metrics

Proposed targets, to be validated with real users.

Metric	Target
Diagnostic completion rate (started to finished)	60% or higher
Users starting a recommended topic within 24h of results	50% or higher
Week-4 retention	25% or higher
Problems solved per active user per week	5 or more
Users rating recommendations "helpful"	80% or higher

14 Milestones

Phase	Deliverable
0	Fix starter: install, lint and build all pass
1	Full DSA graph, seed data, real question bank
2	Scoring, mastery engine, path engine with reasoning
3	Dashboard, topic pages, resources and problems on real data
4	Auth, persistence, RLS review, responsive and accessibility pass
5	Beta with real learners, tune thresholds
6	AI tutor and code review, goals and target dates

15 Risks and Open Questions

Topic	Concern
Question bank	50 questions across 18 areas is about 3 per area, so confidence scores will be noisy. Should the diagnostic adapt to answers or stay fixed?
Mastery formula	Weighting of assessment, practice, recency and decay is undefined.
Practice signals	LeetCode "solved" is self-reported and cannot be verified.
Link rot	External resources break; needs periodic link checks.
Threshold of 4	Requiring 4 on every prerequisite may lock learners out too aggressively. Test alternatives.
AI provider and cost	Undecided. Keep behind the abstraction.
Unverified build	The starter has never completed a production build.

16 Acceptance Criteria

1. npm install, npm run dev, npm run lint and npm run build succeed with no TypeScript errors.
2. Landing page, /learn, DSA path and curriculum render correctly and match the design system.
3. Diagnostic runs, scores per topic and stores mastery.
4. A personalized path is generated with a reason for each recommendation.
5. Dashboard, topic pages, resources and problems display real data.
6. Auth works and progress persists across sessions.
7. RLS blocks access to other users' data.
8. Seed data populates the database with the volumes in section 12.
9. The app is usable on mobile with no horizontal overflow.
10. No obvious console errors.